Exposure at Default Model

CCF Regression for Revolving Retail Credit

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1 What this project does

Exposure at Default (EAD) is the third of the three components in the Basel and IFRS 9 expected-loss identity:

$$\mathrm{EL} = \mathrm{PD} \cdot \mathrm{LGD} \cdot \mathrm{EAD}.$$

For term loans, EAD equals the outstanding balance at default. For revolving facilities (credit cards, overdrafts, lines of credit) the challenge is that the balance at default depends on how much of the undrawn commitment the borrower uses before defaulting. This is modelled via the Credit Conversion Factor (CCF):

$$\mathrm{CCF} = \frac{\mathrm{EAD} - \mathrm{drawn}_{t_0}}{\mathrm{limit} - \mathrm{drawn}_{t_0}}, \qquad \mathrm{EAD} = \mathrm{drawn}_{t_0} + \mathrm{CCF} \cdot (\mathrm{limit} - \mathrm{drawn}_{t_0}).$$

This project builds a CCF scorecard on ∼20,000 revolving retail facilities that default within twelve months, spanning ten origination vintages (2019H1–2023H2). Features are observed at time t_0 (twelve months before default); the target is realised CCF.

2 Data

Each row is one revolving facility observed twelve months prior to default with limit, drawn balance, utilisation, borrower risk score, minimum-payment-ratio behaviour, age, and vintage-level macro context (unemployment, GDP growth).

Product	**Count**	**Mean CCF**	**Std CCF**
Credit card	10 953	0.588	0.181
Overdraft	5 002	0.715	0.166
Line of credit	4 045	0.756	0.162

Table 1: Realised CCF by product type. Overdrafts and lines of credit show higher CCF than credit cards, matching retail credit literature where borrowers with lump-sum facilities draw more of the undrawn commitment before defaulting.

Total EAD across the population ranges from ∼230 EUR to ∼318k EUR, with a median of ∼3.9k EUR and a 99th percentile of ∼87k EUR.

3 Model

Three models are fit and compared on out-of-time data (train 2019H1–2022H2, test 2023H1–2023H2):

- **Fractional logit** (statsmodels GLM Binomial, quasi-likelihood).
- **OLS on logit-transformed CCF** as a linear baseline.
- **XGBoost** on logit-transformed CCF.

Features include log limit, log drawn balance, utilisation at t_0, time to default, risk score at t_0, minimum-payment-ratio, borrower age, and macro variables (unemployment, GDP YoY). Product type is dummy-encoded.

4 Results

4.1 CCF model comparison

Model	MAE	RMSE	R^2	Mean pred	Mean true
Fractional logit	**0.127**	**0.157**	**0.310**	0.646	0.644
OLS-logit	0.128	0.161	0.279	0.666	0.644
XGBoost	0.129	0.161	0.275	0.664	0.644

Table 2: OOT CCF metrics on 4,797 revolving facilities. Fractional logit wins on all three headline metrics. R^2 of $\sim$0.31 reflects the intrinsic difficulty of forecasting individual-borrower draw behaviour twelve months ahead. Mean pred is close to mean true for fractional logit and OLS.

4.2 EAD prediction quality

Once CCF is estimated, EAD follows from the definition. The EAD R^2 is much higher than the CCF R^2 because drawn balance and limit at t_0 are known exactly. Only the incremental draw (CCF $\cdot$ (limit $-$ drawn)) carries model uncertainty.

Model	MAE (EUR)	RMSE (EUR)	R^2	Mean pred	Mean true
Fractional logit	**895**	**2 195**	**0.981**	9 325	9 277
OLS-logit	905	2 305	0.979	9 559	9 277
XGBoost	906	2 316	0.979	9 530	9 277

Table 3: OOT EAD metrics in EUR. Fractional logit is the closest to mean-true EAD (portfolio-level bias of $\sim$0.5%). All three models reach $R^2 > 0.97$ on the EAD scale because most of the variance is explained by the known drawn-at-t_0 and limit inputs.

4.3 CCF calibration

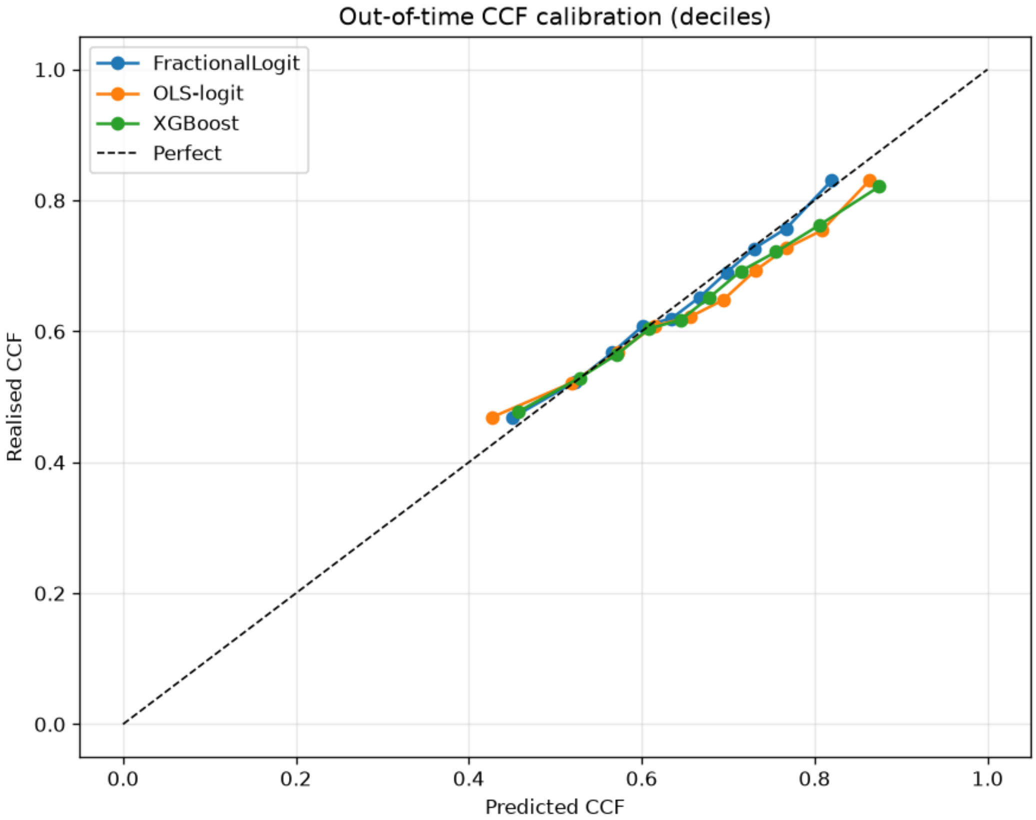

Figure 1: Decile calibration of CCF on the OOT cohort. Fractional logit tracks the diagonal most closely across the range. All three models show slight over-prediction at the highest CCF decile because rare borrowers with CCF $\rightarrow 1$ are hard to distinguish from CCF $= 0.8$ borrowers ahead of default.

4.4 Predicted vs realised EAD

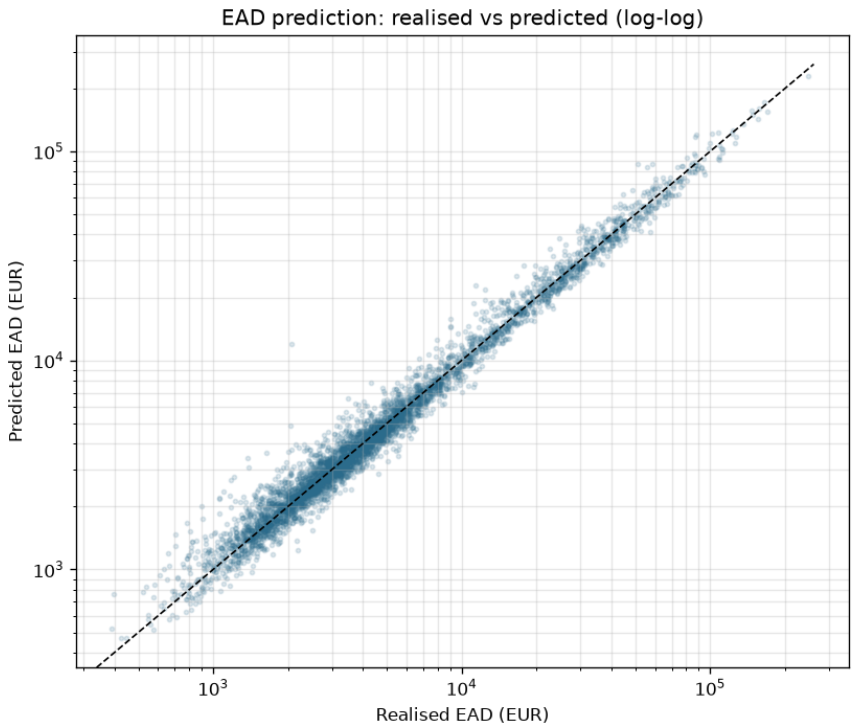

Figure 2: Scatter of predicted versus realised EAD on the OOT cohort (log-log axes). The predictions track the identity line across four orders of magnitude in EAD.

4.5 CCF by product

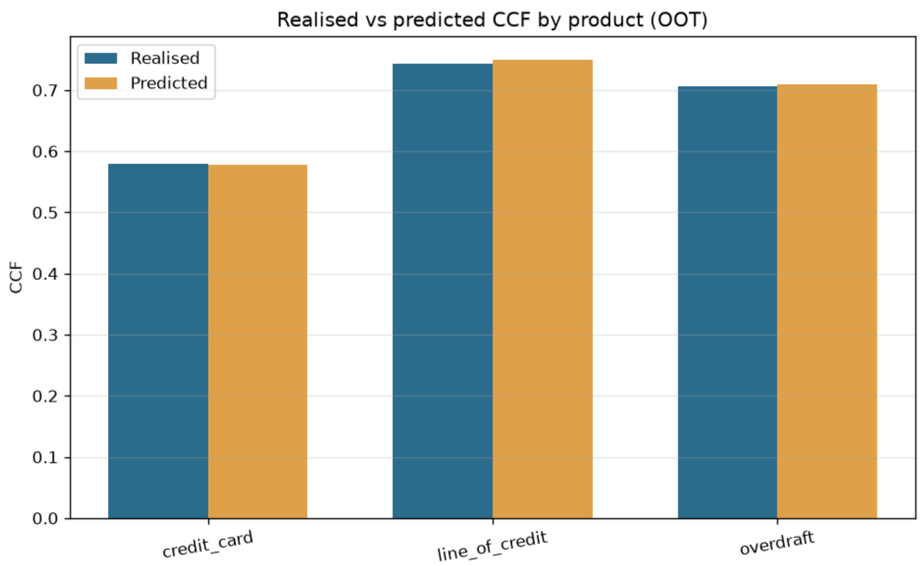

Figure 3: Realised versus predicted mean CCF by product type on the OOT cohort. Model reproduces the product ordering and the level.

4.6 Downturn CCF

Basel IRB requires CCF estimates to be downturn-conservative for regulatory capital. We take the 90th percentile of predicted CCF per product as the downturn point estimate:

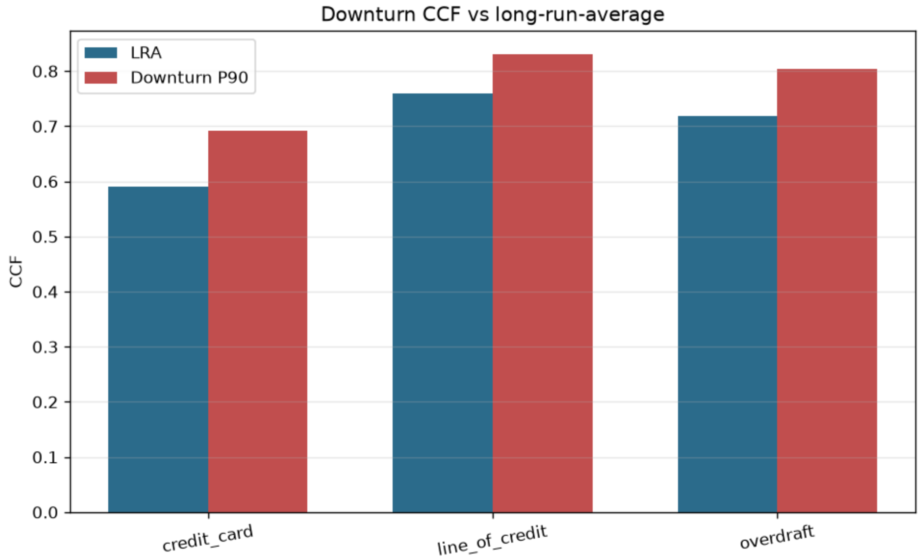

Figure 4: Downturn CCF (P90) versus long-run-average CCF by product. Credit cards show the largest relative uplift because their LRA is lowest, leaving more headroom to draw.

4.7 Feature importance

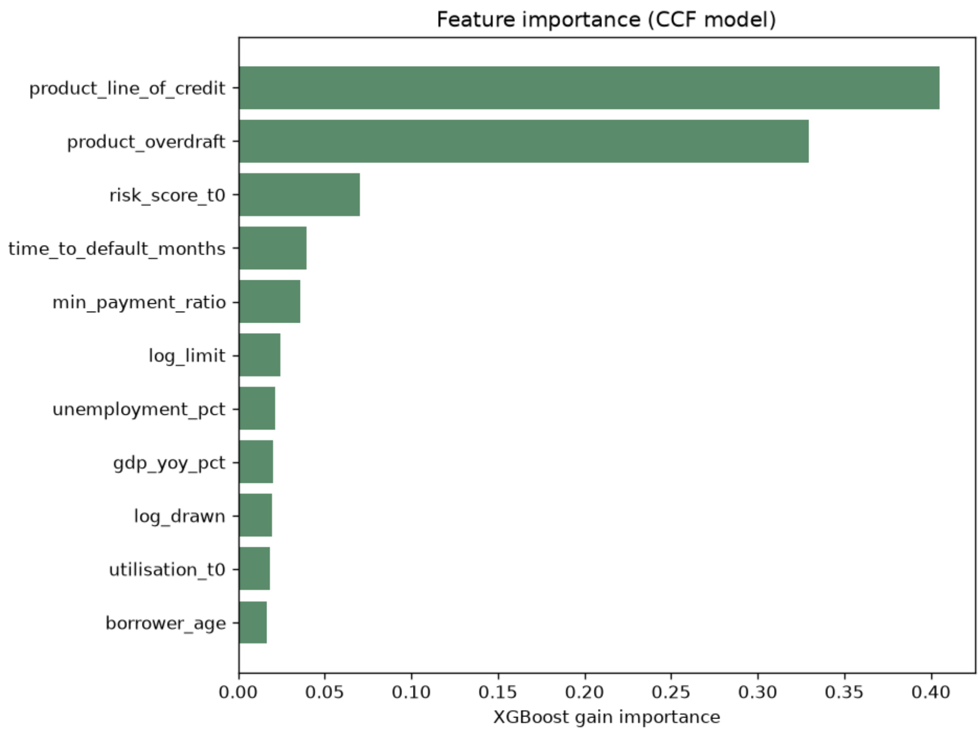

Figure 5: XGBoost gain importance for the CCF model. Risk score at t_0 is the strongest predictor: high-risk borrowers draw more of the undrawn commitment before defaulting. Utilisation at t_0, minimum-payment-ratio, and time to default follow.

4.8 Feature stability (PSI)

Macro-driver PSI is very high between train (2019–2022) and OOT (2023) because the macro window shifted. Idiosyncratic features (limit, drawn, risk score, payment ratio, age) are stable.

Feature	**PSI**
Unemployment pct	17.27
GDP YoY	16.73
Drawn balance	0.010
Credit limit	0.003
Min payment ratio	0.003

Table 4: PSI train vs OOT. As with the LGD model, macro variables should trigger periodic recalibration; idiosyncratic features are stable across time.

5 Basel IRB and IFRS 9 fit

- **Basel IRB EAD requirements:** The scorecard produces product-level CCF estimates with downturn P90 adjustment. EAD for revolving retail is the standard use case for CCF modelling under IRB.
- **IFRS 9 lifetime EAD:** Macro-conditional features let the CCF respond to forward-looking scenarios, which is required for lifetime EAD in stage 2 and 3 exposures.
- **Not fully compliant on:** product-level segmentation may be too coarse. Production models typically further split by originating channel, tenure, and prior behaviour history.

6 Limitations

- Synthetic panel assumes independent facilities. Real portfolios have shared-borrower effects across products.
- CCF horizon is fixed at twelve months. Production would use multiple horizons (up to a full year of the lifetime for IFRS 9).
- Utilisation dynamics are simplified. Real drawing behaviour follows nonlinear paths near limit that a linear or boosted-tree model captures only approximately.
- Behavioural features (minimum-payment-ratio) are static at t_0; in reality they evolve toward default.

7 Files

- `generate_data.py` — synthetic revolving-credit panel
- `train.py` — CCF pipeline (three models, calibration, derived EAD, downturn, PSI)
- `outputs/` — CCF metrics, EAD metrics, PSI, downturn CSVs
- `figures/` — five diagnostic plots